Name: _____

Part1 -Multiple Choice Questions **(45 Points)**

1) $\lim_{x\to\infty} x\ln\left(1+\frac{4}{x}\right)=$

A) 0 B) 4 C) 1 D) ∞

2) Given that $a_1=a_2=1$ and $a_{n+2}=a_{n+1}+a_n$ then $a_7=$

A) 13 B) 12 C) 11 D) 14

3) If $\sum_{k=1}^{\infty} a_k$ is a geometric series, its common ratio $(r=2)$ and $a_8=16$ then $a_6=$

A) 2 B) 4 C) 8 D) 16

4) The range of the function $f(x)=3+2\tanh x$ is

A) $(1,5)$ B) $(-1,1)$ C) $\mathbb{R}$ D) $\mathbb{R}^+$

5) The sequence $\left\{3\sqrt[n]{n}+\left(\frac{2}{5}\right)^n\right\}_{n=1}^{\infty}$

A) converges to 0 B) converges to 3 C) converges to 1 D) diverges

6) The sequence $\left\{n\sin\left(\frac{5}{n}\right)\right\}_{n=1}^{\infty}$

A) converges to 0 B) converges to 1 C) converges to 5 D) diverges

7) The sequence $\left\{\frac{3n+\tan^{-1}n}{n+1}\right\}_{n=1}^{\infty}$

A) converges to 0 B) converges to 2 C) converges to 3 D) diverges

8) Given that $A\cosh 2x+B\sinh 2x=-e^{-2x}$ for any $x\in\mathbb{R}$ then the ordered pair $(A,B)=$

A) $(1,-1)$ B) $(1,1)$ C) $(-1,-1)$ D) $(-1,1)$

9) $\int_0^3 4\tanh 2x\,dx=$

A) $4\ln(\cosh 3)$ B) $4\ln(\cosh 6)$ C) $2\ln(\cosh 3)$ D) $2\ln(\cosh 6)$

10) $\int_0^2 x^2e^x\,dx=$

A) $2e^2-2$ B) $2e^2$ C) e^2-2 D) $2e^2-1$

11) $\int_0^1 4x\ln(x^2+1)\,dx=$

A) $-1+\ln 16$ B) $-2+\ln 4$ C) $-1+\ln 4$ D) $-2+\ln 16$

1) $\lim_{x\to\infty} \dfrac{\ln\left(1+\frac{4}{x}\right)}{1/x} \left(\frac{0}{0}\right) \overset{L.R}{=} \lim_{x\to\infty} \dfrac{-4/x^2}{1+4/x} \cdot -x^2$

$= \lim_{x\to\infty} \left(\dfrac{4}{1+4/x}\right) = 4$ [B]

2) $a_7 = a_{5+2} = a_6 + a_5 = 8+5 = 13$ $(a_n = 5)$

$a_1 = a_2 = 1$, $a_3 = a_2 + a_1 = 2$

$a_4 = a_3 + a_2 = 2+1 = 3$ [A]

$a_5 = a_4 + a_3 = 3+2 = 5$

$a_6 = a_5 + a_4 = 5+3 = 8$

3) $a_8 = 16$, $\dfrac{a_8}{a_7} = 2$ [B]

$a_7 = 8$, $\dfrac{a_7}{a_6} = 2$, $a_6 = 4$

4) $\tanh x = y$

$-1 \le y \le 1 \xrightarrow{\times 2} -2 \le 2y \le 2$ [A]

$\xrightarrow{+3} 1 < 3+2y < 5$, $y \in (1,5)$

5) $\lim_{n\to\infty}\left(3\sqrt[n]{n}+\left(\frac{2}{5}\right)^n\right)=3$ [B]

$\sqrt[n]{n}\to 1$ By L.R $\qquad \left(\frac{2}{5}\right)^n\to 0^{\infty}$ when $a>1$ $=0$

6) $\lim_{n\to\infty}\left(\frac{\sin\left(\frac{5}{n}\right)}{\frac{1}{n}}\right)$ $\overset{w=\frac{1}{n}}{\Longrightarrow}$ $\lim_{w\to 0}\left(\frac{\sin 5w}{w}\right)=5$ [C]

7) $\lim_{n\to\infty}\left(\frac{3n+\tan^{-1}x}{n+1}\right)=3$ (C)

By squeezing Thm:

$$\frac{3n-\frac{\pi}{2}}{n+1}\ \leq\ \frac{3n+\tan^{-1}x}{n+1}\ \leq\ \frac{3n+\frac{\pi}{2}}{n+1}$$

both sides $\to 3$

8) $A\cosh 2x+B\sinh 2x=-e^{-2x}$

$$Ae^{2x}+Ae^{-2x}+Be^{2x}-Be^{-2x}=-2e^{-2x}$$

$$e^{2x}(A+B)+e^{-2x}(A-B)=0e^{2x}-2e^{-2x}$$

1) $A+B=0$

2) $A-B=-2$ $\quad +$

$2A=-2 \longrightarrow A=-1,\ B=1$ $\qquad (-1,1)$ [D]

9) $2\int_0^3 \frac{2\sinh 2x}{\cosh 2x}\,dx = 2\ln(\cosh 2x)\Big]_0^3$

$= 2\ln(\cosh 6) - 2\ln(\cosh 0)$ [D]

$\cosh 0$

$\cosh x = \frac{e^x + e^{-x}}{2}$, $\cosh(0) = \frac{1+1}{2} = 1$

10) $\int_0^2 x^2 e^x\,dx$

$= e^x(x^2 - 2x + 2)\Big]_0^2$

$= e^2(4-4+2) - e^0(2)$ [A]

D	I
x^2	e^x
$2x$	e^x
2	e^x
0	e^x

11) $2\int_0^1 2x\ln(x^2+1)\,dx = 2\int_0^{\ln 2} te^t\,dt$

$t = \ln(x^2+1)$

$e^t = x^2+1$

$e^t\,dt = 2x\,dx$

$= 2e^t(t-1)\Big]_0^{\ln 2}$

$= 4(\ln 2 - 1) - (2)(-1)$

$= 4\ln 2 - 4 + 2 = 4\ln 2 - 2 = \ln 16 - 2$

D	I
t	e^t
1	e^t
0	e^t

[D]

Name: _____ ID: A

12) $\int_0^1 (4\cos^{-1}x - 4\sin^{-1}x)\,dx =$

A) 8 B) 8π C) $8-2\pi$ D) $8+2\pi$

13) Given that $\left(\frac{x^3+x^2+4x+1}{(x^2+4)(x^2+1)} = \frac{1}{x^2+4} + \frac{x}{x^2+1}\right)$ then $\int_0^2 \frac{4x^3+4x^2+16x+4}{(x^2+4)(x^2+1)}\,dx =$

A) $\frac{\pi}{2}+2\ln 5$ B) $\frac{\pi}{4}+2\ln 5$ C) $\frac{\pi}{2}+\ln 5$ D) $\frac{\pi}{4}+\ln 5$

14) The series $\sum_{k=1}^{\infty}\left(\frac{12}{k+1} - \frac{12}{k+3}\right)$

A) converges to 10 B) converges to 5 C) converges to 1 D) diverges

15) If the nth partial sum of the series $\sum_{k=1}^{\infty} a_k$ is $S_n = \frac{12n}{n+1}$ then the series $\sum_{k=4}^{\infty} a_k$

A) converges to 12 B) converges to 3 C) converges to 4 D) diverges

16) The series $\sum_{k=7}^{\infty} \frac{1}{2^{k-5}}$

A) diverges B) converges to 5 C) converges to 1 D) converges to $\frac{1}{2}$

17) The Maclaurin expansion corresponding to the function $f(x) = 2\sin^2 2x$ is

A) $\sum_{k=1}^{\infty}(-1)^k \frac{4^{2k}x^{2k}}{(2k)!}$ C) $\sum_{k=0}^{\infty}(-1)^k \frac{4^{2k}x^{2k}}{(2k)!}$

B) $\sum_{k=0}^{\infty}(-1)^{k+1} \frac{4^{2k}x^{2k}}{(2k)!}$ D) $\sum_{k=1}^{\infty}(-1)^{k+1} \frac{4^{2k}x^{2k}}{(2k)!}$

18) The Taylor expansion (about $x=-2$) corresponding to the function $f(x) = \frac{x+2}{x^2+4x+9}$ is

A) $\sum_{k=0}^{\infty}(-1)^k \frac{(x+2)^{2k}}{5^k}$ C) $\sum_{k=0}^{\infty}(-1)^k \frac{(x+2)^{2k+1}}{5^{k+1}}$

B) $\sum_{k=0}^{\infty}(-1)^k \frac{(x+2)^{2k+1}}{5^k}$ D) $\sum_{k=0}^{\infty}(-1)^k \frac{(x+2)^{2k}}{5^{k+1}}$

12) $4\left(\int_0^1 \cos^{-1} x \, dx - \int_0^1 \sin^{-1} x \, dx \right)$

$$x\cos^{-1}x - \sqrt{1-x^2}\Big]_0^1 - x\sin^{-1}x + \sqrt{1-x^2}\Big]_0^1$$

$$= 4\left(\cos^{-1}(1) - 0 - (-1) - \left(\sin^{-1}1 - 0 - 1 \right) \right)$$

$$= 4\left(1 + 1 - \frac{\pi}{2} \right) = 8 - 2\pi \qquad (C)$$

13) $4\left(\int_0^2 \frac{dx}{x^2+4} + \frac{1}{2}\int_0^2 \frac{2x}{x^2+1}\,dx \right)$

$$2\tan^{-1}\left(\frac{x}{2}\right)\Big]_0^2 + 2\ln(x^2+1)\Big]_0^2$$

$$= 2\tan^{-1}(1) - 2\tan^{-1}0 + 2\ln 5 - 2\ln 1$$

$$= 2\frac{\pi}{4} + 2\ln 5 = \frac{\pi}{2} + 2\ln 5 \qquad (A)$$

14) $12 \sum_{k=1}^{\infty} \frac{1}{k+1} - \frac{1}{k+3}$

$k=1$ $k=2$ $k=3$ $k=4$

$\frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \frac{1}{5} - \frac{1}{7} + \cdots$

$k-1$ k

$+ \frac{1}{k} - \frac{1}{k+2} + \frac{1}{k+1} - \frac{1}{k+3} \Rightarrow \frac{1}{2} + \frac{1}{3} + \frac{1}{k} + \frac{1}{k+1}$

$S = \lim_{k\to\infty} (S_k) = \lim_{k\to\infty} \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{k} + \frac{1}{k+1} \right)$ (0, 0)

$S = 1/2 + 1/3 = \frac{3+2}{6} = \frac{5}{6}$

↝ but remember the sum is multiplied by 12

$12 \cdot S = 12 \frac{5}{6} = 10$ [A]

15) $\sum_{k=1}^{n} a_k$, $S_n = \frac{12n}{n+1}$

find $\sum_{k=4}^{\infty} a_k$ (trick)

$$S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(\frac{12n}{n+1}\right) = 12$$

$$\sum_{k=4}^{\infty} a_k = S - S_3 = S - \left(\frac{12\cdot 3}{4}\right) = 12 - 9 = 3 \quad (B)$$

16) • geo series: $\sum_{n=1}^{\infty} a r^{n-1}$

$$\sum_{n=7}^{\infty} \frac{1}{2^{k-5}} \Rightarrow \sum_{k-5=2}^{\infty} \frac{1}{2^{k-5}} \quad (k-5=n)$$

$$\Rightarrow \sum_{n=2}^{\infty} \frac{1}{2^n} = \sum_{n=0}^{\infty} \frac{1}{2^n} - 1 - \frac{1}{2}$$

$$S = \frac{1}{1-\frac{1}{2}} = 2 \qquad = 2 - 1 - \frac{1}{2} = \frac{1}{2} \quad (D)$$

17) $f(x) = 2\sin^2 2x$ Remember: $\sin^2 x = \frac{1}{2}(1-\cos 2x)$

$f(x) = 1 - \cos 4x$ $\quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

$$1 - \sum_{n=0}^{\infty} \frac{(-1)^n 4^{2n} x^{2n}}{(2n)!} = 1 - \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n 4^{2n} x^{2n}}{(2n)!}\right)$$

متى موجودة بالحيارى

كمان متى لحتى الحيارى

$$= -\sum_{n=1}^{\infty} \frac{(-1)^n 4^{2n} x^{2n}}{(2n)!}$$

بدخل السالب

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 4^{2n} x^{2n}}{(2n)!}$$ [D]

18) $f(x) = \frac{x+2}{(x^2+4x+4)+5} = \frac{x+2}{(x+2)^2+5}$

$= \frac{x+2}{5-(-(x+2)^2)} = \frac{x+2/5}{1-\left(-\left(\frac{x+2}{\sqrt{5}}\right)^2\right)}$, $r = -\left(\frac{x+2}{\sqrt{5}}\right)^2$

$$\frac{x+2}{5}\sum_{k=0}^{\infty}\left(-\left(\frac{x+2}{\sqrt{5}}\right)^2\right)^k = \frac{x+2}{5}\sum_{k=0}^{\infty}\frac{(-1)^k (x+2)^{2k}}{5^k}$$

$$= \sum_{k=0}^{\infty}\frac{(-1)^k (x+2)^{2k+1}}{5^{k+1}}$$ [C]

Name: _____

Part1 -Multiple Choice Questions **(45 Points)**

1. $\lim_{x\to\infty} x\ln\left(1+\frac{5}{x}\right)=$

A. 5 B. 0 C. 1 D. ∞

2. Given that $a_1=a_2=1$ and $a_{n+2}=a_{n+1}+a_n$ then $a_6=$

A. 5 B. 6 C. 7 D. 8

3. If $\sum_{k=1}^{\infty} a_k$ is a geometric series, its common ratio $(r=2)$, and $a_8=16$ then $a_{10}=$

A. 16 B. 32 C. 64 D. 128

4. The range of the function $f(x)=1+2\tanh x$ is

A. $(-1,1)$ B. $(-1,3)$ C. $\mathbb{R}$ D. $\mathbb{R}^+$

5. The sequence $\left\{2\sqrt[n]{n}+\left(\frac{3}{5}\right)^n\right\}_{n=1}^{\infty}$

A. converges to 2 B. converges to 0 C. converges to 1 D. diverges

6. The sequence $\left\{n\sin\left[\frac{4}{n}\right]\right\}_{n=1}^{\infty}$

A. converges to 4 B. converges to 0 C. converges to 1 D. diverges

7. The sequence $\left\{\frac{2n+\tan^{-1}n}{n+1}\right\}_{n=1}^{\infty}$

A. converges to 0 B. converges to 2 C. converges to 3 D. diverges

8. Given that $A\cosh 2x+B\sinh 2x=e^{-2x}$ for any $x\in\mathbb{R}$ then the ordered pair $(A,B)=$

A. $(1,-1)$ B. $(1,1)$ C. $(-1,1)$ D. $(-1,-1)$

9. $\int_0^3 8\tanh 2x\,dx=$

A. $4\ln(\cosh 3)$ B. $4\ln(\cosh 6)$ C. $2\ln(\cosh 3)$ D. $2\ln(\cosh 6)$

10. $\int_0^1 x^2e^x\,dx=$

A. $e-2$ B. e C. $2e$ D. $e+2$

11. $\int_0^1 2x\ln(x^2+1)\,dx=$

A. $-1+\ln 16$ B. $-2+\ln 4$ C. $-1+\ln 4$ D. $-2+\ln 16$

1) $\lim_{x\to\infty} \dfrac{\ln\left(1+\frac{5}{x}\right)}{1/x} \left(\frac{0}{0}\right) \overset{L.R}{=} \lim_{x\to\infty} \dfrac{-5/x^2}{1+5/x} \cdot -x^2$

$= \lim_{x\to\infty} \left(\dfrac{5}{1+5/x}\right) = 5$ (A)

$(\ln 5)$

2) $a_6 = a_{4+2} = a_5 + a_4 = 5+3 = 8$

$a_1 = a_2 = 1$, $a_3 = a_2 + a_1 = 2$

$a_4 = a_3 + a_2 = 2+1 = 3$ (D)

$a_5 = a_4 + a_3 = 3+2 = 5$

3) $a_8 = 16$, $\dfrac{a_9}{a_8} = 2$ (C)

$a_9 = 32$, $\dfrac{a_{10}}{a_9} = 2$, $a_{10} = 64$

4) $\tanh x = y$

$-1 \le y \le 1 \longrightarrow -2 \le 2y \le 2$ (B)

$-2+1 \le 1+2y \le 2+1$, $y \in (-1,3)$

5) $\lim_{n\to\infty}\left(2\sqrt[n]{n}+\left(\frac{3}{5}\right)^n\right)=2$ [A]

$\sqrt[n]{n}\to 1$ By L.R $\qquad \left(\frac{3}{5}\right)^n\to a^{\infty}$ when $a>1 = 0$

6) $\lim_{n\to\infty}\left(\frac{\sin\left(\frac{4}{n}\right)}{\frac{1}{n}}\right)$ $\xrightarrow{w=\frac{1}{n}}$ $\lim_{w\to 0}\left(\frac{\sin 4w}{w}\right)=4$ [A]

7) $\lim_{n\to\infty}\left(\frac{2n+\tan^{-1}x}{n+1}\right)=2$ (B)

By squeezing Thm: $\frac{2n-\frac{\pi}{2}}{n+1}\to 2 \le \frac{2n+\tan^{-1}x}{n+1} \le \frac{2n+\frac{\pi}{2}}{n+1}\to 2$

8) $A\cosh 2x + B\sinh 2x = e^{-2x}$

$$Ae^{2x}+Ae^{-2x}+Be^{2x}-Be^{-2x}=2e^{-2x}$$

$$e^{2x}(A+B)+e^{-2x}(A-B)=0e^{2x}+2e^{-2x}$$

1) $A+B=0$

2) $A-B=2$ $\quad (1,-1)$ [A]

$2A=2 \longrightarrow A=1,\ B=-1$

9) $4\int_0^3 2\frac{\sinh 2x}{\cosh 2x}\,dx = 4\ln(\cosh 2x)\Big]_0^3$

$= 4\ln(\cosh 6) - 4\ln(\cosh 0)$ [B]

$\cosh 0 = 1$

$\cosh x = \frac{e^x + e^{-x}}{2}$, $\cosh(0) = \frac{1+1}{2} = 1$

10) $\int_0^1 x^2 e^x\,dx$

D	I
x^2	e^x
$2x$	e^x
2	e^x
0	e^x

$= e^x(x^2 - 2x + 2)\Big]_0^1$

$= e^1(1 - 2 + 2) - e^0(2)$ [A]

11) $\int_0^1 2x\ln(x^2+1)\,dx = \int_0^{\ln 2} te^t\,dt$

D	I
t	e^t
1	e^t
0	e^t

$t = \ln(x^2+1)$

$e^t = x^2 + 1$

$e^t\,dt = 2x\,dx$

$= e^t(t-1)\Big]_0^{\ln 2}$

$= 2(\ln 2 - 1) - (-1)$

$= 2\ln 2 - 1 = -1 + \ln 4$

[C]

Name: _____

12. $\int_0^1 (2\cos^{-1}x - 2\sin^{-1}x)\,dx =$

A. 4 B. 4π C. $4+\pi$ D. $4-\pi$

13. Given that $\left(\frac{x^3+x^2+4x+1}{(x^2+4)(x^2+1)} = \frac{1}{x^2+4} + \frac{x}{x^2+1}\right)$ then $\int_0^2 \frac{2x^3+2x^2+8x+2}{(x^2+4)(x^2+1)}\,dx =$

A. $\frac{\pi}{2}+2\ln 5$ B. $\frac{\pi}{4}+2\ln 5$ C. $\frac{\pi}{2}+\ln 5$ D. $\frac{\pi}{4}+\ln 5$

14. The series $\sum_{k=1}^{\infty}\left(\frac{6}{k+1} - \frac{6}{k+3}\right)$

A. converges to 10 B. converges to 5 C. converges to 1 D. diverges

15. If the nth partial sum of the series $\sum_{k=1}^{\infty} a_k$ is $S_n = \frac{12n}{n+1}$ then the series $\sum_{k=3}^{\infty} a_k$

A. converges to 12 B. converges to 3 C. converges to 4 D. diverges

16. The series $\sum_{k=6}^{\infty} \frac{1}{2^{k-5}}$

A. converges to 2 B. converges to $\frac{1}{2}$ C. converges to 1 D. diverges

17. The Maclaurin expansion corresponding to the function $f(x) = 2\sin^2 3x$ is

A. $\sum_{k=1}^{\infty}(-1)^k \frac{6^{2k}x^{2k}}{(2k)!}$ C. $\sum_{k=0}^{\infty}(-1)^k \frac{6^{2k}x^{2k}}{(2k)!}$

B. $\sum_{k=1}^{\infty}(-1)^{k+1} \frac{6^{2k}x^{2k}}{(2k)!}$ D. $\sum_{k=0}^{\infty}(-1)^{k+1} \frac{6^{2k}x^{2k}}{(2k)!}$

18. The Taylor expansion (about $x=-2$) corresponding to the function $f(x) = \frac{x+2}{x^2+4x+7}$ is

A. $\sum_{k=0}^{\infty}(-1)^k \frac{(x+2)^{2k}}{3^k}$ C. $\sum_{k=0}^{\infty}(-1)^k \frac{(x+2)^{2k}}{3^{k+1}}$

B. $\sum_{k=0}^{\infty}(-1)^k \frac{(x+2)^{2k+1}}{3^k}$ D. $\sum_{k=0}^{\infty}(-1)^k \frac{(x+2)^{2k+1}}{3^{k+1}}$

12) $2\left(\int_0^1 \cos^{-1}x\,dx - \int_0^1 \sin^{-1}x\,dx\right)$

$$x\cos^{-1}x - \sqrt{1-x^2}\Big]_0^1 - x\sin^{-1}x + \sqrt{1-x^2}\Big]_0^1$$

$$= 2\left(\cos^{-1}(1) - 0 - (-1) - \left(\sin^{-1}1 - 0 - 1\right)\right)$$

$$= 2\left(1+1-\frac{\pi}{2}\right) = 4-\pi \qquad (D)$$

13) $$2\int_0^2 \frac{dx}{x^2+4} + \int_0^2 \frac{2x}{x^2+1}\,dx$$

$$2\tan^{-1}\left(\frac{x}{2}\right)\Big]_0^2 + \ln(x^2+1)\Big]_0^2$$

$$= 2\tan^{-1}(1) - 2\tan^{-1}0 + \ln 5 - \ln 1$$

$$= 2\frac{\pi}{4} + \ln 5 = \frac{\pi}{2} + \ln 5 \qquad [C]$$

14) $6\sum_{k=1}^{\infty} \frac{1}{k+1} - \frac{1}{k+3}$

$k=1$ $k=2$ $k=3$ $k=4$

$\frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \frac{1}{5} - \frac{1}{7} + \cdots$

$k-1$ k

$+ \frac{1}{k} - \frac{1}{k+2} + \frac{1}{k+1} - \frac{1}{k+3} \Rightarrow \frac{1}{2} + \frac{1}{3} + \frac{1}{k} + \frac{1}{k+1}$

$S = \lim_{k\to\infty} (S_k) = \lim_{k\to\infty} \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{k} + \frac{1}{k+1} \right)$ (6, 6)

$S = 1/2 + 1/3 = \frac{3+2}{6} = \frac{5}{6}$

↝ but remember the sum is multiplied by 12

$6 \cdot S = 6 \frac{5}{6} = 5$ [B]

15) $\sum_{k=1}^{n} a_k$, $S_n = \frac{12n}{n+1}$

find $\sum_{k=3}^{\infty} a_k$ (trick)

$$S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(\frac{12n}{n+1}\right) = 12$$

$$\sum_{k=3}^{\infty} a_k = S - S_2 = S - \left(\frac{\cancel{12}^{4}\cdot 2}{\cancel{3}}\right) = 12 - 8 = 4 \quad [C]$$

16) • geo series: $\sum_{n=1}^{\infty} a r^{n-1}$ or $\sum_{n=0}^{\infty} a r^{n}$

$$\sum_{k=6}^{\infty} \quad k-5 \Rightarrow \sum_{k-5=1}^{\infty} \frac{1}{2^{k-5}} \quad k-5=n \Rightarrow \sum_{n=1}^{\infty} \frac{1}{2^n}$$

$$= \underbrace{\sum_{n=0}^{\infty} \frac{1}{2^n}}_{\text{1}} - 1 = 2 - 1 = 1 \quad [C]$$

$$\hookrightarrow S = \frac{a}{1-r} = \frac{1}{1-1/2} = 2$$

17) $f(x) = 2\sin^2 3x$ Remember: $\sin^2 x = \frac{1}{2}(1-\cos 2x)$

$f(x) = 1 - \cos 6x$ $\quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

$$1 - \sum_{n=0}^{\infty} \frac{(-1)^n 6^{2n} x^{2n}}{(2n)!} = 1 - \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n 6^{2n} x^{2n}}{(2n)!}\right)$$

منشوفو جوده بالجنارال

كمان منشوفو بالجنارال

$$= -\sum_{n=1}^{\infty} \frac{(-1)^n 6^{2n} x^{2n}}{(2n)!}$$

بنخذ السالب

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 6^{2n} x^{2n}}{(2n)!}$$ [B]

18) $f(x) = \frac{x+2}{(x^2+4x+4)+3} = \frac{x+2}{(x+2)^2+3}$

$= \frac{x+2}{3-(-(x+2)^2)} = \frac{x+2/3}{1-\left(-\left(\frac{x+2}{\sqrt{3}}\right)^2\right)}$, $r = -\left(\frac{x+2}{\sqrt{3}}\right)^2$

$$\frac{x+2}{3}\sum_{k=0}^{\infty}\left(-\left(\frac{x+2}{\sqrt{3}}\right)^2\right)^k = \frac{x+2}{3}\sum_{k=0}^{\infty}\frac{(-1)^k (x+2)^{2k}}{3^k}$$

$$= \sum_{k=0}^{\infty}\frac{(-1)^k (x+2)^{2k+1}}{3^{k+1}}$$

[D]

Part 2: Essay Question (5 Points)

Find the interval and radius of convergence of the following power series

$$\sum_{k=1}^{\infty}(-1)^k \frac{5^k(x-7)^k}{k!} \qquad c=7$$

interval: $x \in \{7\}$, Radius: $R=0$

By ratio test:

$$\lim_{k\to\infty}\left|\frac{5^{k+1}(x-7)^{k+1}}{(k+1)!}\cdot\frac{k!}{5^k(x-7)^k}\right|$$

$$=\lim_{k\to\infty}\left|\frac{5^k\,5\,(x-7)^k(x-7)}{(k+1)\,k!}\cdot\frac{k!}{5^k(x-7)^k}\right|$$

$$=|x-7|\lim_{k\to\infty}\left(\frac{5}{k+1}\right)<1$$

$$|x-7|\cdot\infty<1$$

$x\neq 7$: ∞

$x=7$: $0<1$